

Product Requirements Document (PRD): Gallo Bottling Sanitation Hub

1. Project Overview & Objective

The **Gallo Bottling Sanitation Hub** is a digital solution designed to replace manual, paper-based sanitation checklists for the Modesto bottling room lines (specifically focusing initially on lines like 91, 92, and 93). The primary objective is to transition to a simple, **mobile-first web application** that captures checklist entries, automatic timestamps, and photos. The system will store records centrally and support a **real-time dashboard for Sanitation, Quality, Maintenance, and Operations leadership** to improve communication, reduce manual scanning, and ensure audit readiness.

2. Target Audience & User Personas

- **Sanitation Technician (Operator):** Floor workers who perform the deep cleans, typically using **Surface Pro tablets**. They require a highly usable interface optimized for **gloved hands**. Operators can fill out checklists but are **not allowed to edit the master templates**.
- **Administrator (Team Lead / Quality Assurance):** Leadership personnel who typically access the system via **desktop web browsers**. They need functionality to create, edit, or remove checklist items, review submitted work, flag problematic entries, and **approve the final sanitation status**.

3. Functional Requirements

A. Digital Checklist Execution

- **Line Selection:** Operators must be able to select their specific bottling line (e.g., Line 91, 92, or 93).
- **Task Evaluation:** Operators must be able to mark tasks as **Complete, Incomplete, or N/A**.
- **Incomplete Item Logic:** If an operator marks a task as "Incomplete" (failure found), the system **must prompt them to add a mandatory explanatory note or attach a photo**.
- **Partial Submissions (Drafts):** Operators must have the ability to save an uncompleted checklist as a draft so it can be handed off to another operator on the next shift to finish.
- **Automatic Timestamps:** The system must capture **start and end timestamps for each different equipment group automatically**. Manual timestamp entry is not permitted.

B. Admin Live Dashboard

- **Review & Approval:** Admins must be able to view submitted checklists, separate reviewed from unreviewed checklists, and approve them.

- **List View & Previews:** The dashboard should load checklists in a list format of **20 per page, with a preview of the selected checklist to the right.**
- **Data Filtering:** Admins must be able to filter submissions by date, line, time, or employee names/identification numbers.
- **Summary & Metrics:** The system must generate an automated deep clean summary that calculates **exactly how long the deep clean took to complete.**
- **Audit Export:** Admins must be able to **export checklist data to PDF** for audit readiness.

C. Security & Authentication

- **Single Sign-On (SSO):** The system will use **Entra ID** for authentication. Eventually, employees may log in using their badge readers.
- **Accountability:** The system must record the names and timestamps of whoever is editing a document.
- **Access Control:** The system must lock out users after consecutive failed logins and require **re-authentication after periods of inactivity.**

D. Machine Learning Classifier (Proposed Feature)

- **Automated Data Entry:** An operator can upload a photo of the equipment.
- **Classification:** The ML model will classify the required checklist entries and autofill them.
- **Validation:** The operator must manually scroll, confirm all autofilled entries are correct, and then upload the entry.

4. Non-Functional Requirements

- **Usability:** The UI must be a mobile-first web app that is **operable with gloved hands.**
- **Scalability:** The architecture must be scalable to support approximately **5 production sites and ~2,000 employees.**
- **Availability:** Bottling lines run **24/7**, requiring high system uptime. If a submission fails due to network issues, data must be retained locally for a retry.
- **Compliance:** Sanitation records must be retained for **2 years** to comply with food safety regulations.

5. Technology Stack & Infrastructure

- **Frontend:** React (v18), Vite, TypeScript, and Redux Toolkit with Material UI [18, 35, conversation history].
- **Backend:** Serverless architecture using **AWS API Gateway and AWS Lambda (Node.js).**
- **Database & Storage:** **Aurora Serverless PostgreSQL** (relational data), **DynamoDB** (NoSQL for flexible checklist templates), and **Amazon S3** (for photos and PDF exports).
- **Authentication:** Amazon Cognito for Operator/Admin role management.

- **Infrastructure:** Self-hosted on AWS, provisioned using **AWS CloudFormation**, and monitored via CloudWatch and X-Ray. Expected modest production AWS spend is \$200–\$600 USD/month.